



HMIET- I Automatic Voltage Regulator

自动电压调整器



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Preface

Introduction

This guide and the relevant operating or service manual documentation for the equipment provide full information on safe handling, commissioning and testing of this equipment.

Documentation for equipment ordered from Huaming Equipment CO., LTD is dispatched separately from manufactured goods and may not be received at the same time. Therefore, this guide is provided to ensure that printed information normally present on equipment is fully understood by the recipient.

Before carrying out any work on the equipment, the user should be familiar with the contents of this manual, and read relevant chapter carefully.

This chapter describes the safety precautions recommended when using the equipment. Before installing and using the equipment, this chapter must be thoroughly read and understood.

Health and Safety

The information in this chapter of the equipment documentation is intended to ensure that equipment is properly installed and handled in order to maintain it in a safe condition.

When electrical equipment is in operation, dangerous voltages will be present in certain parts of the equipment. Failure to observe warning notices, incorrect use, or improper use may endanger personnel and equipment and cause personal injury or physical damage.

Before working in the terminal strip area, the equipment must be isolated.

Proper and safe operation of the equipment depends on appropriate shipping and handling, proper storage, installation and commissioning, and on careful operation, maintenance and servicing. For this reason, only qualified personnel may work on or operate the equipment.

Qualified personnel are individuals who:

Are familiar with the installation, commissioning, and operation of the equipment and of the system to which it is being connected;

Are able to safely perform switching operations in accordance with accepted safety engineering practices and are authorized to energize and de-energize equipment and to isolate, ground, and label it;

Are trained in the care and use of safety apparatus in accordance with safety engineering practices;

Are trained in emergency procedures (first aid).

Instructions and Warnings



damage could occur if safety precautions are disregarded.

CAUTION! means that light personal injury or equipment damage may occur if safety precautions are disregarded.

NOTICE! Is particularly applies to damage to device and to resulting damage of the protected equipment.

**DANGER!**

NEVER allow a open current transformer (CT) secondary circuit connected to this device while the primary system is live. Open CT circuit will produce a dangerously high voltage that cause death.

**WARNING!**

ONLY qualified personnel should work on or in the vicinity of this device. This personnel MUST be familiar with all safety regulations and service procedures described in this manual. During operating of electrical device, certain part of the device is under high voltage. Severe personal injury and significant device damage could result from improper behavior.

**WARNING!**

Do NOT touch the exposed terminals of this device while the power supply is on. The generated high voltage causes death, injury, and device damage.

**WARNING!**

Thirty seconds is NECESSARY for discharging the voltage. Hazardous voltage can be present in the DC circuit just after switching off the DC power supply.

**CAUTION!**

- Earthing

Securely earthed the earthing terminal of the device.

- Operating environment

ONLY use the device within the range of ambient environment and in an environment free of abnormal vibration.

- Ratings

Check the input ratings BEFORE applying AC voltage/current and power supply to the device.

- Printed circuit module

Do NOT attach or remove printed circuit board if the device is powered on.

- External circuit

Check the supply voltage used when connecting the device output contacts to external circuits, in order to prevent overheating.

- **Connection cable**

Carefully handle connection cables without applying excessive force.

NOTICE!

HMIET - I Automatic Voltage Regulator

The firmware may be upgraded to add new features or enhance/modify existing features, please **MAKE SURE** that the version of this manual is compatible with the product in your hand.

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1. Introduction

1.1 Scope of Application

HMIET-I Automatic Voltage Regulator (Hereinafter referred to as HMIET-I or AVR) is an intelligent device to raise or lower the on-load tap changer position by designed automatic logic or manual operation, therefore adjust the transformer voltage.

Main Features:

- Line voltage, current and frequency measurement, active current and reactive current calculation;
- OLTC Position monitoring via BCD or 4-20mA signals;
- Configurable input and output signals;
- Configurable blocking for overvoltage, undervoltage, overcurrent, etc;
- Abnormal locking of OLTC status;
- In manual mode, raise or lower the switch gear through buttons, input signals, or remote control signals;
- Adjust the position according to the target voltage or reactive current in automatic mode;
- Supports 3 configurable and selectable reference voltages
- Three reference voltages can be switched through buttons, input signals, or remote control signals;
- The device can operate independently or in parallel (up to 6 devices can be connected in parallel), and in parallel mode, gear adjustment can be carried out through master-slave mode or minimum loop method;
- Configurable voltage drop compensation parameters;
- Scalable to up to 3 sets of CT and VT monitoring
- Special application voltage regulation supporting split transformers
- Support voltage regulation for special applications of high, medium, and low winding transformers
- Customized automatic voltage regulation solutions can be developed according to special project requirements;
- Support IEC61850 or Modbus RTU protocols communication with SCADA;
- SNTP clock synchronization;

1.2 Hardware Configuration

- HMIET-I full-width of 19-inch 4U cabinet for easy installation;
- HMIET-I device is equipped with a 320*240 LCD screen on the front panel, 10 LED indicators and a 9-key keyboard;

- HMIET-I device provides 4 channels of 4-20mA input measurement; 1 channel of voltage, 1 channel of current measurement;
- HMIET-I device provides 32 channels of input indications and 24 channels of output indication;
- HMIET-I device provides 2 RS485 serial ports, 2 100M optical or 2 RJ45 electrical Ethernet interfaces , and supports communication protocols such as IEC61850, Modbus etc.

1.3 Ordering Notice

- The conventional configuration of the input plug-in voltage level is DV110V, and DC220V can be customized;
- Expanded RS485 communication need to be confirmed;
- The functional requirements or extended functions of special projects should be determined by technical personnel;
- With the improvement and enhancement of the product, Huaming reserves the right to make changes to the provided technical data and usage instructions

2. Technical Parameters

2.1 Electrical Parameters

2.1.1 Power Supply

Standard	GB/T 8367-1987 (IEC 60255-11:2008)
Input range	85-265V AC/DC two-way
Total power	<15W

2.1.2 Switching value input

Standard	IEC 60255-1:2009	
Rated voltage	110V DC	220V DC
Starting voltage	60.5V~77V	121V~154V
Rated current	1.1mA	1.1mA
Maximum current	300Vdc	
Withstand voltage level	2000Vac, 2800Vdc	

2.1.3 4~20mA DC current measurement

Range	4mA-20mA
Load resistance	75 Ω
Measurement error	$\pm 0.2\%$ Full Range

2.1.4 AC value measurement

Current measurement error	Level 0.5
Measuring range	0~10 A
Voltage measurement error	Level 0.5
Measuring range	0~400 V

2.2 Environment Parameters

Standard	GB/T 14047-1993 (idt IEC 60225-1:2009)
Operating temperature	-20°C~+70°C
Storage and temperature	-30°C~+80°C
Relative humidity	5%~95%, no condensation or freezing inside the device

2.3 Communication Interface

2.3.1 Rs-485 interface

Bit rate	4.8kbit/s、9.6kbit/s、19.2kbit/s
Communication protocol	Modbus
Maximum load	32

Transmission distance	<500 m
Connecting type	Shielded twisted pair

2.3.2 Ethernet Interface

Connecting type	RJ-45 or fiber-optical Ethernet LC interface, multimode, wavelength 1310nm
Transmission rate	100Mbits/s, self-adaptive
Transmission standard	10Base-T/100Base-TX, 100base-FX
Communication protocol	IEC 61850
Connecting type	Category 5 shielded network cable or multimode fiber

2.4 Type Tests

2.4.1 Environment test

Low temperature test	GB/T 2423.1-2008 Low temperature: 2h at -20°C
High temperature test	GB/T 2423.2-2008 High temperature: 2h at +55°C
Alternating damp heat test	GB/T 2423.4-2008 High temperature: 55°C, Low temperature: 25°C, RH 95%, 2 loops of 12h+12h

2.4.2 Mechanical test

Vibration test	GB/T 11287-2000 level 1
Impact test	GB/T 14537-1993 level 1
Crash test	GB/T 14537-1993 level 1

2.4.3 Electrical insulation test

Insulation voltage	GB 14598.27-2008, 2kV, 50Hz, 1 min
Insulation resistance	GB 14598.3-2006, >100MΩ at 500Vdc
Impulse voltage	GB 14598.3-2006, 5KV
Overvoltage category	GB 14598.3-2006, category III

2.4.4 Electromagnetic compatibility test

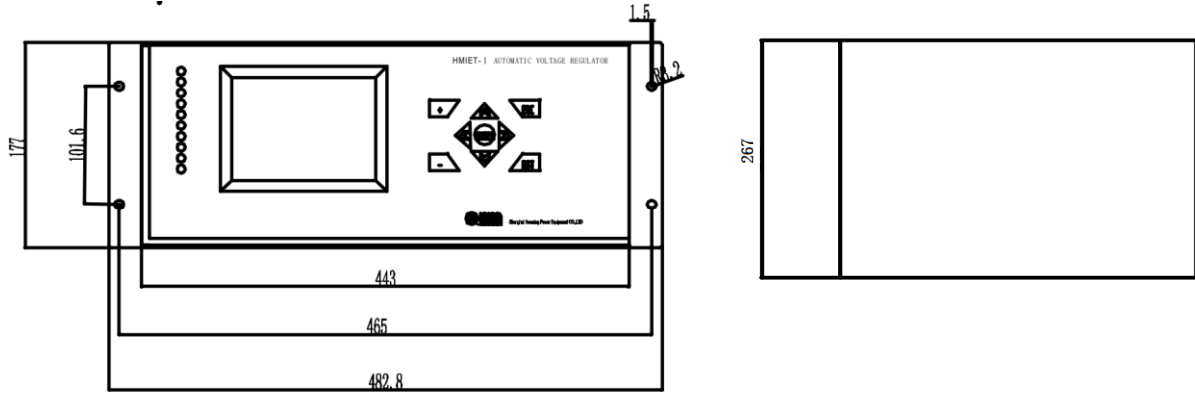
1MHz burst interference test	Level 3 (GB/T 17626-12-1998)
Electrostatic discharge test	Level 4 (GB/T 14598.14-2010)
Radiated electromagnetic field interference test	Level 3 (GB/T 17626.3-2006)
Fast transient interference test	Level A (GB/T 14598.10-2007)
Surge test	Level 4 (GB/T 17626.5-2008)
Radio frequency conducted interference test	Level 3 (GB/T 17626.6-2008)
Power frequency magnetic	Level 5 (GB/T 17626.8-2006)

field immunity test	
Pulsed magnetic field test	Level 5 (GB/T 17626.9-1998)
Damped oscillating magnetic field test	Level 5 (GB/T 17626.10-1998)
The auxiliary power port voltage drops, Short interruption	When it drops to 40% of the rated voltage, the device can be guaranteed to work normally within 300ms; short-term interruption: the device will not restart within 100ms

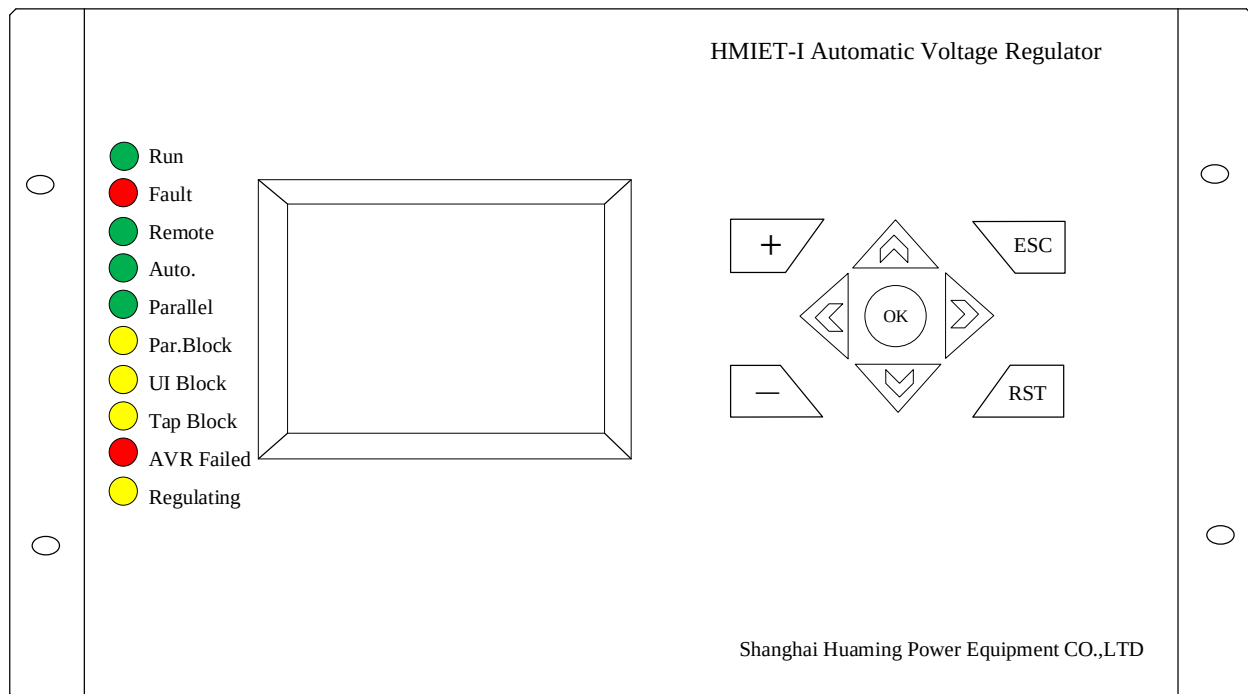
3. Hardware Description

The device adopts all aluminum chassis design, surface wire drawing treatment and aluminum shell oxidation. The board is installed by rear plugging, and the front panel is provided with LCD, LED indicator and keyboard.

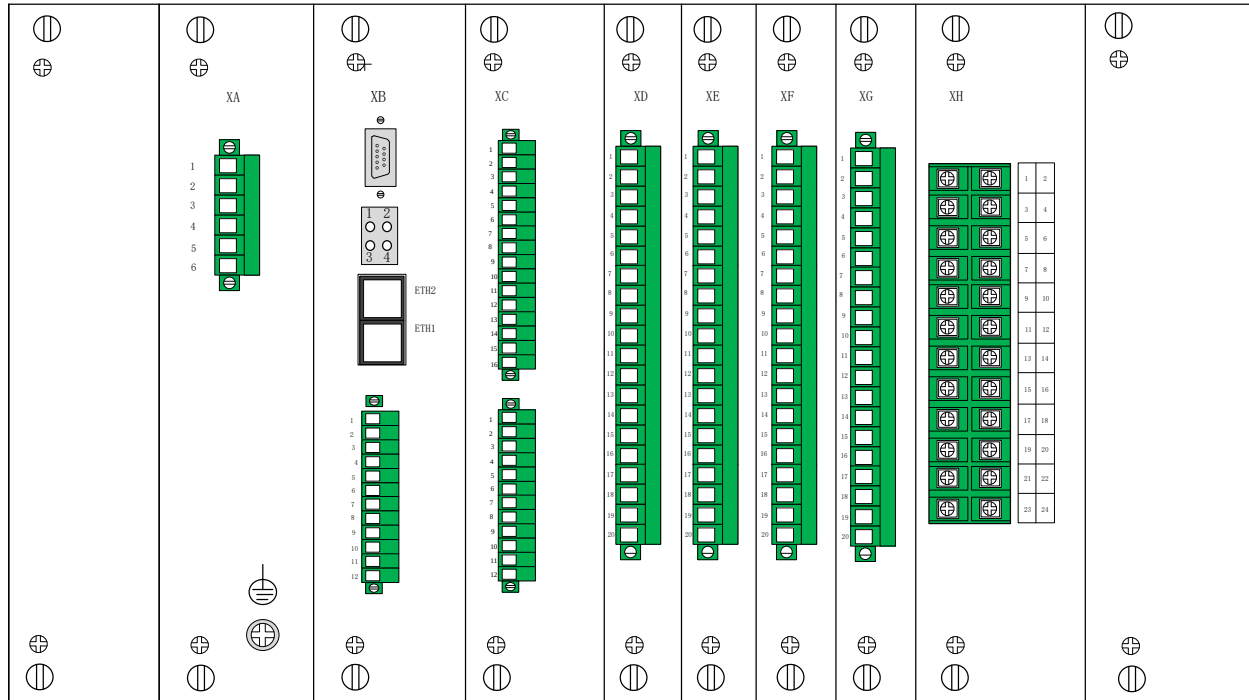
3.1 Device structure size



3.2 Front Panel



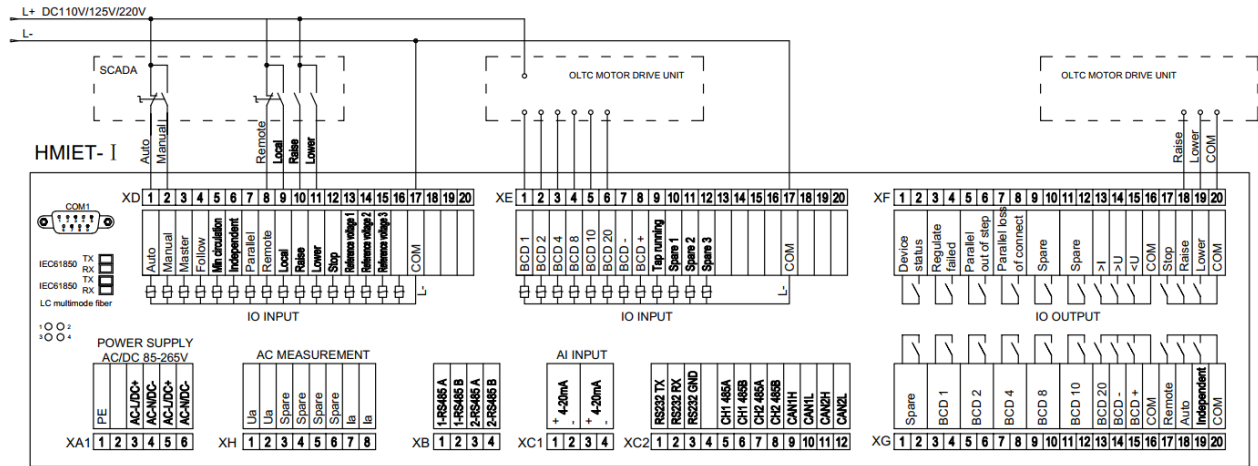
3.3 Terminal Definition and Description



Device backplane terminal definition diagram

The hardware configuration of the device (from left to right):

- Power supply board;
- Communication board;
- Main control board;
- Input board 1;
- Input board 2;
- Output board 1;
- Output board 2;
- AC collection board.



3.3.1 Auxiliary Power Board

Terminal group XA is a six core power terminal that includes two power inputs, supports AC/DC redundancy, and has an input range of 50/60Hz 85-265V AC/DC.

XA	
1	POWER GND
2	NC
3	POWER 1 AC L / DC +
4	POWER 1 AC N /DC -
5	POWER 2 AC L / DC +
6	POWER 2 AC N /DC -

3.3.2 Communication Board

XB is a communication board, including a DB9 serial port debugging port, a set of LED indicators, 2 Ethernet (optical port, electrical port optional), at most 8 RS485 interfaces (1 channel is optional for IRIG-B timing interface).

XB	
1	Rs485_A1
2	Rs485_B1
3	Rs485_A2
4	Rs485_B2

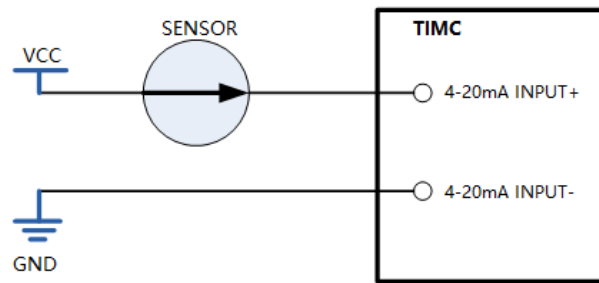
3.3.3 Sampling Board

The sampling board contains 2 channels of 4-20mA analog signal. XC1 terminal collects 2 4-20mA analog signals:

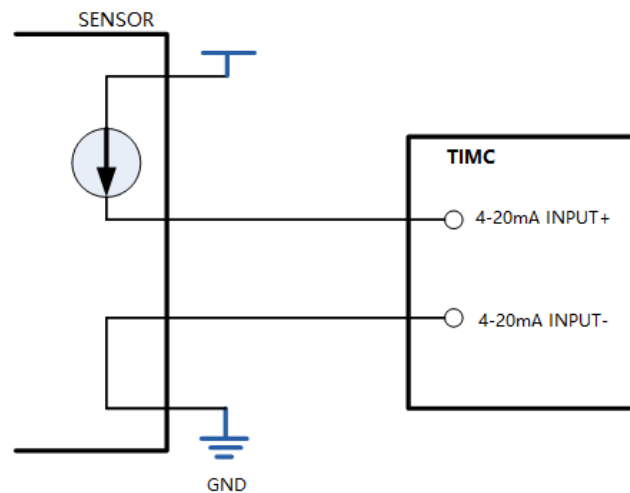
XC1 端子定义	
1	4~20mA_1+
2	4~20mA_1-

3	4~20mA_2+
4	4~20mA_2-

The 4-20mA analog signal acquisition terminal of the device supports two wiring methods of two-wire and four-wire sensors. The wiring description is shown in the figure below:



Two-wire sensor wiring diagram



Four-wire sensor wiring diagram

The Rs232 debugging port of XC2 can be connected to a computer to view detailed device operation messages and update upgrade programs. The Rs485 interface can transmit data to upper layers using the Modbus protocol. CAN bus is a parallel communication bus used for multiple AVR devices in parallel;

XC2			
1	Rs 232 Tx	7	Rs 485 _A
2	Rs 232 Rx	8	Rs 485 _B
3	GND	9	CAN_1 H
4	Null	10	CAN_1 L

3.3.4 Input Boards

The switch input board can be equipped with two input voltages, DC110V or DC220V, and each board supports 16 input signals.

XD			
1	Auto mode	11	Down
2	Manual mode	12	Stop
3	Master mode	13	Reference voltage1
4	Follow mode	14	Reference voltage2
5	Min circulation	15	Reference voltage3
6	Independent	16	NULL
7	Parallel	17	COM
8	Remote control	18	NULL
9	Local control	19	NULL
10	Raise	20	NULL
XE			
1	BCD 1	11	Spare
2	BCD 2	12	Spare
3	BCD 4	13	Spare
4	BCD 8	14	NULL
5	BCD 10	15	NULL
6	BCD 20	16	NULL
7	BCD -	17	COM
8	BCD +	18	NULL
9	Motor work signal	19	NULL
10	Latching input	20	NULL

3.3.5 Output Boards

Each board supports 12 dry node output signals, which can be configured as normally open or normally closed nodes according to requirements. The contact capacity is AC250V, 5A or DC30, 5A, and the breaking time is less than 10ms.

XF			
1	Device status	11	Overvoltage lock
2	COM	12	COM
3	Regulate failed	13	Undervoltage lock
4	COM	14	Circulation latching
5	Parallel loss of step	15	Spare
6	COM	16	COM

7	Parallel loss of connect	17	Stop
8	COM	18	Raise
9	Overcurrent lock	19	Down
10	COM	20	COM
XG			
1	Spare	11	BCD 10
2	COM	12	COM
3	BCD 1	13	BCD 20
4	COM	14	BCD -
5	BCD 2	15	BCD +
6	COM	16	COM
7	BCD 4	17	Remote
8	COM	18	Auto
9	BCD 8	19	Independent
10	COM	20	COM

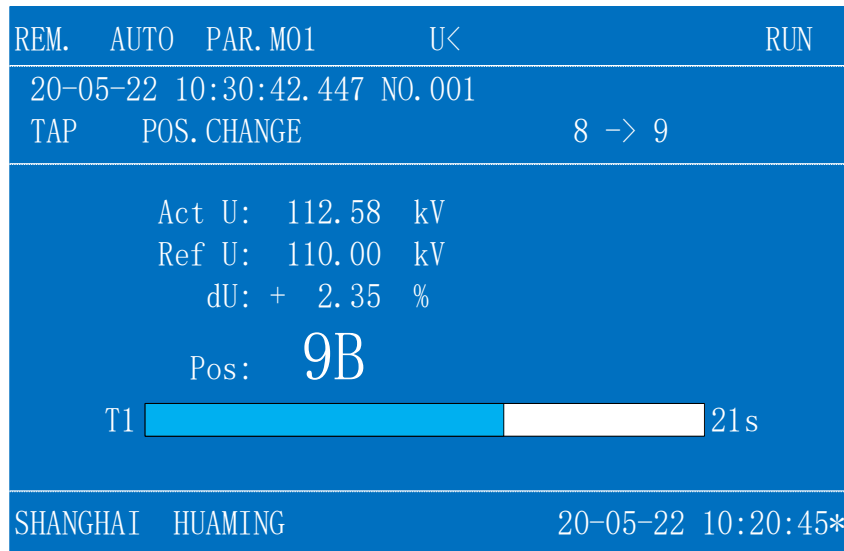
3.3.6 AC Board

The AC measurement board provides 1 circuit of current/voltage measurement, which can be expanded to 3 circuit;

XH		
1	U	2
3	SPARE	4
5	SPARE	6
7	I	8
9	SPARE	10
11	SPARE	12

4. Operation Manual

4.1 Main Window



The main window includes four parts information:

4.1.1 Work mode

Remote: displays "REM." The operation mode of AVR at this time is in remote control mode, accepting remote input signals or communication protocol operations. The "Remote" LED is always on.

Local: displays "LOC." HMIET-I operation is only available via device HMI. The "Remote" LED is always off.

Auto: displays "AUTO" , The AVR automatically controls voltage regulation based on the reference voltage setting. The "Auto" LED is always on.

Manual: displays "HAND" , HMIET-I operates OLTC according to external signals(communication or digital indications) or HMI operation .The "Auto" LED is always off.

Parallel: displays "PAR." . HMIET-I works in parallel mode. The "Parallel" LED is always on.

Independent: displays "INDEPEND" , HMIET-I works in Independent mode. The "Parallel" LED is always off.

Parallel Master: displays such as "M01" in first column means HMIET-I works in parallel master mode and the address is 1.

Parallel Follower: displays such as "F02" in first column means HMIET-I works in parallel follower mode and the address is 2.

Parallrl Circ. Current: displays such as "PAR.CIR.C" in first column means HMIET-I works in parallel circulation current mode.

Parallel Block: While several AVRs work in parallel mode, the master AVR checks the parallel setting and connection and step difference. "SETERR" means the AVRs settings are not matched such as the count, the master or follower role, the local or remote mode.

"CONNERR" means any follower in parallel loses connection with master.

"STEPERR" means step difference between follower and master is more than the setting setp limits. Any parallel error will block the parallel raising and lowering operations and "Par.Block" LED lights on.

U/I Block: "U<" in first column means under voltage block; "U>" means over voltage block. "I>" means over current block. In Automatic mode, any blocking will disable AVR function and "U/I Block" LED lights on.

Regulating: displays "↑" in first column means tap is raising, "↓" means that tap is lowering. "Regulating" LED lights on while raising or lowering.

Debug mode: "DBG" indicates that the device is working in debug mode, user can operate parameter setting, debugging equipment, etc .. "Run" LED flashes every second.

Run mode: "RUN" indicates that the device is working in normal mode, All enabled logics work properly and device control commands can be operated in real time. "Run" LED lights on always.

Device Fault: Device always do self check while running, self check error or inner communication error will cause device fault. "DERR" indicates device working in debugging mode and having device fault. "RERR" indicates device working in running mode and having device fault. Device fault blocks AVR to operate tap raising or lowering and "Fault" LED lights on.

4.1.2 System Events

The new events are pushed to the second column of main window, showing the time, composition, event type, and result of the event. The RESET key returns all the event information in this area.

4.1.3 RT measurements

Display real-time voltage and reference voltage of transformer primary or secondary sides, and tap changer position and percent of voltage difference.

The process bar pops up while device is in automatic regulation case, the number shows the left seconds to activate regulation.

4.1.4 System Info

Display system time and company information, If there is a NTP clock synchronization

successfully, display “*” after the system time.

4.2 Keyborad Operation

Press “UP” or “DOWN” or “LEFT” or “RIGHT” key to move the cursor;

Press “+” or “-” to change the digit;

Press “Esc” to cancel operation;

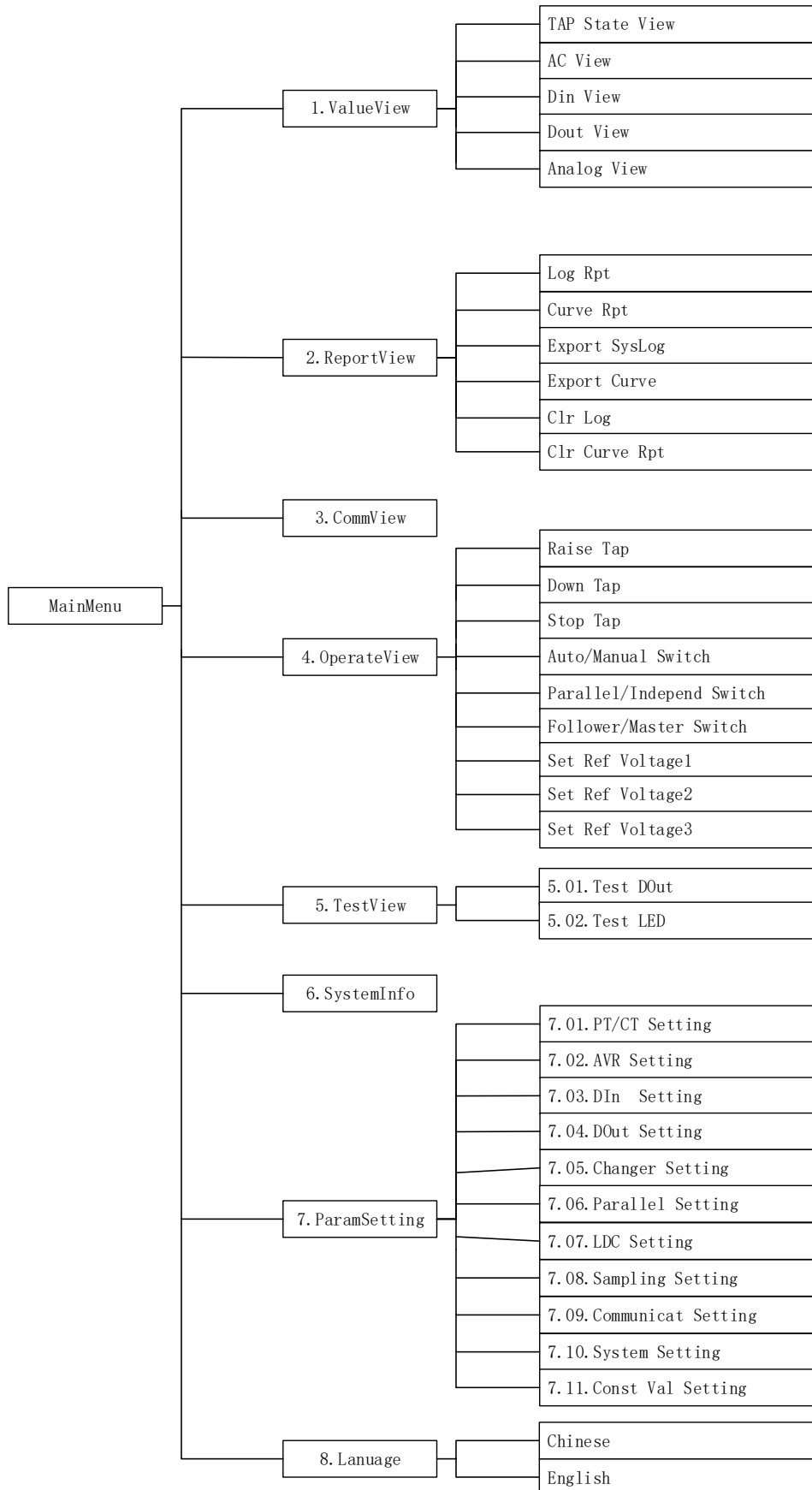
Press “OK” to confirm operation.

4.2.1 Switch “Run” to “Debug” mode

- To check current working mode is “RUN” ;
- Press both “UP” and “DOWN” keys meanwhile ;
- Enter correct 4 digits password (default:1233) ;
- To check current working mode is “DBG” .

LOC.	HAND	INDEPEND	DBG
.07	PASSWORD		
[	0- 9999]		
	1233		
SHANGHAI HUAMING		20-05-22 10:20:45	

4.3 Main Menu



4.4 Test Value Display

4.4.1 OLTC Status

REM.	AUTO	PAR. M01	RUN
TAP	STEP		10
TAP	POS		9
TAP	INFO		2
ACT	COUNTER		5430
IND:BCD01			ON
IND:BCD02			OFF
IND:BCD03			OFF
IND:BCD04			ON
IND:BCD05			OFF
IND:BCD06			OFF
SHANGHAI	HUAMING		20-05-22 10:20:45

This page displays tap real-time data such as step, position, tap union information, tap operation counter and tap related state signals. The signals includes inner indications and alarms which can be configured to the HMIET-I physical digital inputs and outputs in menu "7.03. DIIn Setting" and "7.04. DOut Setting" .

4.4.2 AC Display

REM.	AUTO	PAR. M01	RUN
U=	100.00 V		110.00 kV
I=	1.000 A		500.0 A
Φ =	0 °		
Ia=	1.000 A		
Ir=	0.000 A		
F=	50.00 Hz		
SHANGHAI	HUAMING		20-05-22 10:20:45

This page shows the measured voltage, current, angle difference, active current, reactive current and frequency, the left side is the secondary value, the right side is the primary value. The primary and secondary value of PT and CT should be configured in menu "7.01.PT/CT Setting" .

4.4.3 Input View

REM.	AUTO	PAR.	M01	RUN			
DI:1	2	3	4	5	6	7	8
[1]:	0	0	0	0	0	0	0
[2]:	0	0	1	1	0	1	0
[3]:	0	0	0	0	0	0	0
[4]:	0	0	0	0	0	0	0
SHANGHAI HUAMING				20-05-22 10:20:45			

This page displays the real-time state of the 32 channel input signals of the device . "1" indicates that there is an input signal and "0" indicates that there is no input signal.

4.4.4 Output View

REM.	AUTO	PAR.	M01	RUN			
DO:1	2	3	4	5	6	7	8
[1]:	0	0	0	0	0	0	0
[2]:	0	0	1	1	0	1	0
[3]:	0	0	0	0	0	0	0
SHANGHAI HUAMING				20-05-22 10:20:45			

This page displays the real-time state of the 24 channel output signals of the device . "1" indicates that there is an output signal and "0" indicates that there is no output signal.

4.4.5 Analog View

REM.	AUTO	PAR. MO1	RUN
ANG1=	6.01	mA	45.5 °C
ANG2=	0.00	mA	0.00 mA
ANG3=	0.00	mA	0.00 mA
ANG4=	0.00	mA	0.00 mA
ANG5=	0.00	mA	0.00 mA
ANG6=	0.00	mA	0.00 mA
ANG7=	0.00	mA	0.00 mA
ANG8=	0.00	mA	0.00 mA
SHANGHAI HUAMING			20-05-22 10:20:45

This page displays 8 channels of 4-20mA analog signals from the device. The left side displays real-time measured values of 4-20mA, and the right side displays the actual converted values.

4.5 Report View

Users can view event reports and curve reports(not available), export report records, clear report records under this menu. Event composition includes device self-test event, tap change event, input signal change event, output signal change event, operation setting event and so on.

REM.	AUTO	PAR. MO1	RUN
.2.1	Log	Rpt	
.2.2	Curve	Rpt	
.2.3	Export	SysLog	
.2.4	Export	Curve	
.2.5	Clr	Log	
.2.6	Clr	Curve Rpt	
SHANGHAI HUAMING			20-05-22 10:20:45

4.6 Communication View

REM.	AUTO	PAR. M01	RUN
PORT	STATE	PROTOCOL	
Net_1	OK	IEC61850 Server	
Net_2	OK	IEC61850 Server	
485_1	ERR	SHM-D Master	
485_2	–	NONE	
485_3	–	NONE	
485_4	–	NONE	
SHANGHAI HUAMING			20-05-22 10:20:45

In this page user can view the communication status of the protocols configured on the network port and RS485 serial port of the device ." OK" means that the communication is normal;" ERR" means the communication is abnormal;"-" means that the interface is not configured to communicate.

4.7 Operation View

REM.	HAND	PAR.	M01	RUN
. 4. 1	Raise		TAP	
. 4. 2	Lower		TAP	
. 4. 3	Stop		TAP	
. 4. 4	Auto/Manual Switch			
. 4. 5	Parallel/Independ Switch			
. 4. 6	Follower/Master Switch			
. 4. 7	Set Ref. Voltage 1			
. 4. 8	Set Ref. Voltage 2			
. 4. 9	Set Ref. Voltage 3			
SHANGHAI HUAMING				20-05-22 10:20:45

In **local mode** user can raise ,lower or stop OLTC , switch AVR mode and set reference voltage in HMI. Mode switching operations includes auto or manual mode switching ,parallel or independence mode switching, master or follower mode switching. All operations need to be confirmed via correct system password.

Please be in mind that the tap operation is only allowed in **local manual mode** and the AVR must be master if device works in parallel mode.

4.8 Test View

User can test if the Output relays and LED can work properly, and the test needs to do in debugging mode.

		RUN
.5.1.1	SET ODD DOUTs ON	
.5.1.2	SET EVEN DOUTs ON	
.5.1.3	SET ALL DOUTs OFF	
SHANGHAI HUAMING		20-05-22 10:20:45

4.9 Parameter Setting

All parameter settings need to be set in **debug mode**, effective in running mode.

4.9.1 PT/CT Setting

		DBG
.03	PrimaryVoltage	
[	10.0~ 1000.0]	
	110.0kV	
Primary vlotage in kV		
SHANGHAI HUAMING		20-05-22 10:20:45

User can set the primary and secondary values of PT and CT. Before the equipment is shipped from the factory, it will be configured based on the information provided by the project. If there are different parameters, they can be easily changed on-site

4.9.2 AVR Setting

User can set the parameters regarding automatic voltage regulation such as the referenced primary voltage, bandwidth voltage(percent), time delay T1 (T2 is not necessary). Over voltage blocking and under voltage blocking and over load current blocking settings are all set in this menu.

The AVR function will be invalid while any blocking case is activated.

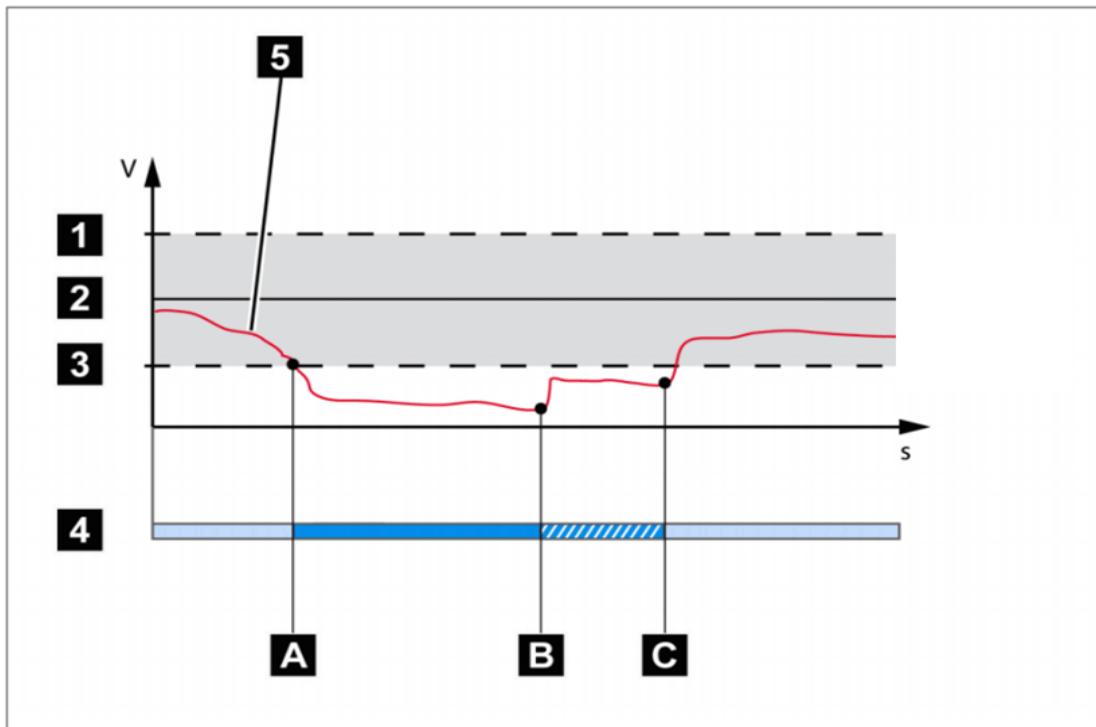
REM.	HAND	PAR. M01	DBG
.04	BINDWIDTH		
[	0.50	~ 9.00]	
		2.50%	
Bind width voltage(percent)			
SHANGHAI	HUAMING	20-05-22	10:20:45

Band width is a percent of reference voltage, while act voltage is out of bind width it will cause T1 or T2 timer working, if the act voltage backs to the bind width T1 timer will roll back until to 0, otherwise AVR will be tripped while T1 or T2 is completed. T2 should always be set lesser than T1.

Setting an appropriate bandwidth requires to know the voltage changed value of each tap step of OLTC, in the case of SHZV-type OLTC, which has a tap position from 1 to 17 and an adjustable voltage range of $\pm 10\%$, so the adjustable voltage for each tap step is:

$$\Delta U = (20\% \cdot U_{ref}) / 16 = 1.25\% \cdot U_{ref}$$

In order to avoid excessively low bandwidth value leading to frequently raising and lowering operations, B% should be greater than $0.6 \cdot \Delta U$. For Tap Regulation logic, B% should be lesser than once ΔU .



1: Bind Width +B%

2: **Reference Voltage**

3: Bind Width -B%

4: Time Line

5: Act Voltage

A: Act U is out of range, T1 start

B: T1 is completed. Trips tap changing

C: T2 is completed. Trips tap changing

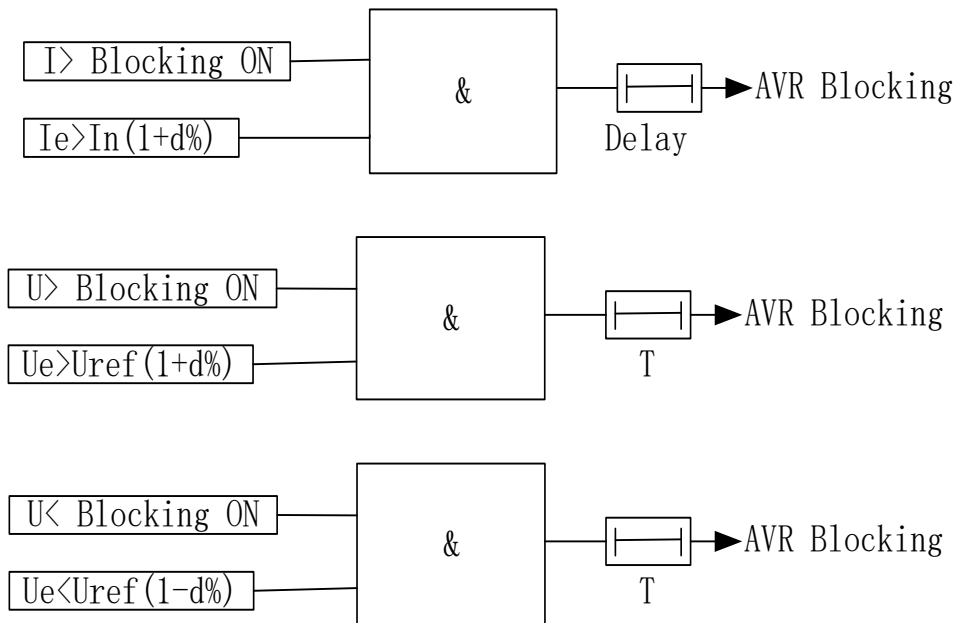
Example:

The reference voltage is 33.0kV and B% is 1.00% ,The ΔU between current measured voltage and reference voltage causes T1 timer counting. The main HMI shows the T1 timer is counting and AVR will trip tap changing from position 10 to 9B after T1 timeout.

REM.	AUTO	PAR. MO1	RUN
20-05-22	10:30:42.447	NO.001	
OUTPUT	AUTO MODE	D08 ON	
Act U: 33.45 kV Ref U: 33.00 kV dU: + 1.36 % Pos: 10 T1 2s			
SHANGHAI	HUAMING	20-05-22 10:20:45	

REM.	AUTO	PAR. MO1	RUN
20-05-22	10:30:57.432	NO.001	
TAP	POS.CHANGE	10 -> 9	
Act U: 33.04 kV Ref U: 33.00 kV dU: + 0.12 % Pos: 9B			
SHANGHAI	HUAMING	20-05-22 10:20:45	

The over voltage blocking logic works while "U > BLOCKING" is set ON, similarly set "U < BLOCKING" and "I > BLOCKING". The logical schemas are as followed:



REM.	HAND	INDEPEND	DBG
.07	AVR	DIRECTION	
[	0	~	1]
1:		REVERSE	
AVR	DIRECTION		
SHANGHAI	HUAMING	20-05-22	10:20:45

Raise step, select NORMAL for voltage rise and REVERSE for voltage drop.

4.9.3 Input Setting (Factory configured)

The state, alarm and other input information supported by the device are mapped to the physical signal. Any input signal can be mapped to XD or XE connections according to the design. The default value of 0 is undefined mapping.

The order input indications 1 to 16 correspond XD01 to XD16; The order input indications 17 to 32 correspond XE01 to XE16.

The OLTC position can be represented by BCD, 4-20mA, binary code, or Gray code. Communication connection can also be used. If it is a communication connection, any blocking signal will generate a total blocking signal

4.9.4 Output Setting

The state, alarm and other output information supported by the device are mapped to the physical signal. Any output signal can be mapped to DOUT1 to DOUT24 according to the design. The default value of 0 is undefined mapping.

The order DOUT1 to DOUT12 correspond XF connections; The order DOUT13 to DOUT24 correspond XG connections;

REM. HAND INDEPEND	DBG
.07 Raise	
[0 ~ 24]	
1: DOUT01	
Raise Output Signal	
SHANGHAI HUAMING	
20-05-22 10:20:45	

There are 49 indications or alarms can be configured to any one of the 32 digital outputs. Tap position can be indicated via BCD, BIN or Gray codes. Multiple alarms can be configured to a same digital output. No.9 "SELF CHECK ERROR" means AVR device has faulty, on the contrary No.10 "SELF CHECK SUCCESS" means device is OK. By communication way, any OLTC block will generate "TC GENERAL BLOCK" if communication is OK, otherwise "TC COMM FAILED" will be generated if communication is failed.

Example:

INDEPEND MODE use XF15-XF16 output:

REM. HAND INDEPEND	DBG
.37 INDEPEND MODE	
[0 ~ 24]	
9: XF15-XF16	
INDEPEND MODE	
SHANGHAI HUAMING	
20-05-22 10:20:45	

STOP use XG18-XG20 output:

REM.	HAND	INDEPEND	DBG
. 31	STOP		
[	0 ~ 24]		
23:	XG18-XG20		
STOP	SIGNAL		
SHANGHAI	HUAMING	20-05-22	10:20:45

4.9.5 Tap Changer Setting

The Tap parameter includes the tap type setting, the tap maximum and minimum position setting, the tap operation number, the input and output tap signal coding type setting and so on.

“Connect mode” means the connection mode between HMIET-I device and OLTC .

“Communication” option is adapt to Huaming SHM-D 、 SHM-X OLTC as so far. If

“Communication” is selected use XC2-5 and XC2-6 to connect through the serial port.

Remote or **Local** control mode can be set while device works in **RUN mode** via HMI or DIN. Designing a switch in panel which connects AVR digital input is convenient for user to change remote or local control mode.

	DBG
. 04	MANUAL CONTROL MODE
[	0 ~ 1]
1:	LOCAL
Manual	Control Mode
SHANGHAI	HUAMING
20-05-22	10:20:45

Tap position signal can support BCD code, binary code , Gray code or 4-20mA signal. If 4-20mA signal is selected, in .7.8.2 Analog Setting ,a minimum (4mA) and a maximum (20mA) tap position value should be set correctly.

	DBG
.04 INPUT TAP CODE TYPE	
[0~ 10]	
0: BCD	
Input tap position in code type	
SHANGHAI HUAMING	20-05-22 10:20:45

Tap change counter shows the current tap operation number, it starts counting from the number which is set newly.

	DBG
.08 TAP CHANGE COUNT	
[0~ 9999999]	
0,002,238	
Tap operation number	
SHANGHAI HUAMING	20-05-22 10:20:45

4.9.6 Parallel Setting

HMIET-I device supports to regulate at most 6 AVR in parallel mode. Default mode of AVR device is follower. Before transformers paralleling operation user should finished parallel settings such as the number of devices in parallel, the device unique Id, the maximum allowed step difference. The device works in parallel or independence mode, takes as master or follower, which is switchable via HMI or external hardwire signal or communication.

REM. HAND PAR. M01	RUN
. 01 PARALLEL UNBLOCK	
[0~ 1]	
<u>0</u> : OFF	
ParallelUnblock	
SHANGHAI HUAMING	
20-05-22 10:20:45	

While “PARALLEL UNBLOCK” is set ON, it means that the automatic working device can manually raise or lower tap changer temporarily which is like in manual mode.

	DBG
. 03 PARALLEL METHOD	
[0~ 2]	
<u>1</u> : MASTER	
ParallelMethod	
SHANGHAI HUAMING	
20-05-22 10:20:45	

Paralleled AVR's exchange information via canbus communication. Anytime anyone AVR has block such as parallel setting error, lost connection, out of step, over current, over voltage, under voltage, OLTC block, OLTC communication faulty, AVR faulty, the parallel raising or lowering operation will be block. A reminder is all parallel AVR's should be work in same local or remote mode.

Normally master AVR decides to raise or lower tap and the followers just to follow raising or lowering. Huaming recommends the allowed step difference is 0 in parallel OLTCs.

LOC.	AUTO	PAR. CIR. C	DBG
.05	PARALLEL	ADDRESS	
[	0~	6]	
	<u>1</u>		
ParallelAddress			
SHANGHAI HUAMING			
20-05-22 10:20:45			

The address of the master device is 1, and the addresses of other devices increase in ascending order.

LOC.	AUTO	PAR. CIR. C	DBG
.06	CIR. CCURRENT	BANDWITH	
[	0.0~	2000.0]	
	2000.0	A	
CircCurrentBandwidth			
SHANGHAI HUAMING			
20-05-22 10:20:45			

LOC.	AUTO	PAR. CIR. C	DBG
.07	CIR. CCURRENT	LIMITION	
[	0.0 ~	3000.0]	
	3000.0	A	
CircCurrentLimition			
SHANGHAI HUAMING			
20-05-22 10:20:45			

The Circulation current bandwidth needs to be measured. the tap position and the circulation current for each transformer are noted down, the circulation current change per tap can then be determined from the noted values. Circulation current band must to be

4.9.8 AC Coeff. Setting

The accuracy coefficient of communication volume is usually set during the factory debugging of the device and directly put into use on site

REM. HAND PAR. M01	DBG
.01 PU	
[500~2000]	PU=100.01 V
1000	
SHANGHAI HUAMING	20-05-22 10:20:45

4.9.9 Analog Coeff. Setting

The accuracy coefficient of analog signals is usually calibrated and set using a signal generator during factory debugging, and is directly put into use on site

	DBG
.01 ACCURACY	
[900~ 1100]	ANG1= 8.00mA
1000	
Coff. Setting	
SHANGHAI HUAMING	20-05-22 10:20:45

Each analog channel can be set the corresponding values of 4mA and 20mA which are used to calculate the actual signal value.

	DBG
.02 MINNUM	
[-999.999~ 999.999]	
000.000	
4mA Value	
SHANGHAI HUAMING	20-05-22 10:20:45

4.9.10 Communication Setting

HMIET-I supports 2 network interfaces and 2 RS485 interfaces to communicate with external device or system, the communicated parameters and protocols can be set independently. HMIET-I supports NTP protocol to synchronize clock, the NTP server IP address can be set in network setting items. Only one NTP server can be used for two network ports. HMIET-I supports network protocol like Modbus Tcp and IEC61850 etc. HMIET-I supports serial protocol like Modbus Rtu and CDT etc .

REM. HAND PAR. M01	DBG
.05 PROTOCOL	
[0 ~ 1]	
1: IEC61850 Server	
Network interface protocol	
SHANGHAI HUAMING	20-05-22 10:20:45

For IEC61850 communication with SCADA, AVR provides "HMIET01.icd" file which IEDName is "HMIET001" ("001" is the last three digits of your IP1 address).

Attention:

SCADA controls AVR to raise or lower tap changer while AVR is working in **run mode** and **remote mode** without any block. SCADA sets AVR settings while AVR is working in **debug** mode.

4.9.11 System Setting

System setting includes the system clock setting, system password setting ,LCD light time setting and so on. The system password is four digital number(**default 1233**). It is a common key when user switching device state,setting parameters and clear system logs and other operations. When any key is pushed, LCD backlight will be on until the setting time is over,but if the setting is 0, it means the backlight will always be on.

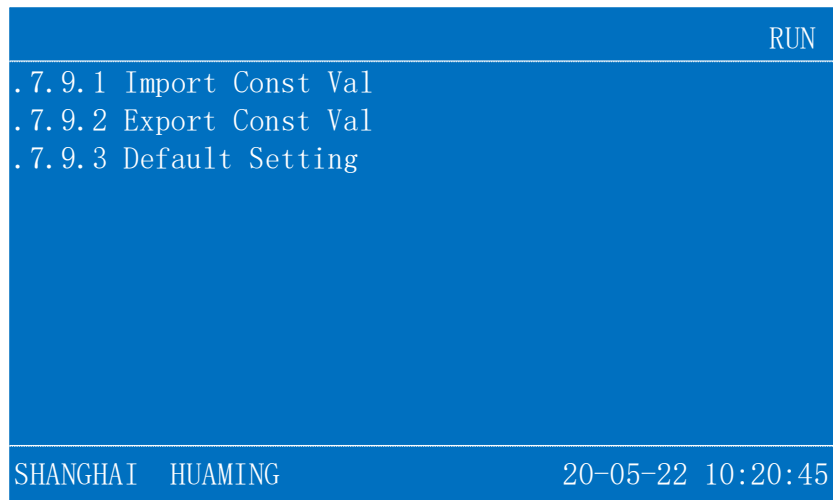
REM. HAND PAR. M01	DBG
. 02 PASSWORD	
[0 ~ 9999]	
1111	
System password	
SHANGHAI HUAMING	
20-05-22 10:20:45	

Enter the time zone and frequency as required.

REM. HAND PAR. M01	DBG
. 05 TIME ZONE	
[-24.00~ 24.00]	
8.00	
TimeZone	
SHANGHAI HUAMING	
20-05-22 10:20:45	

REM. HAND PAR. M01	DBG
. 06 FREQ	
[0 ~ 1]	
0: 50HZ	
Frequency	
SHANGHAI HUAMING	
20-05-22 10:20:45	

4.9.12 Const Val setting



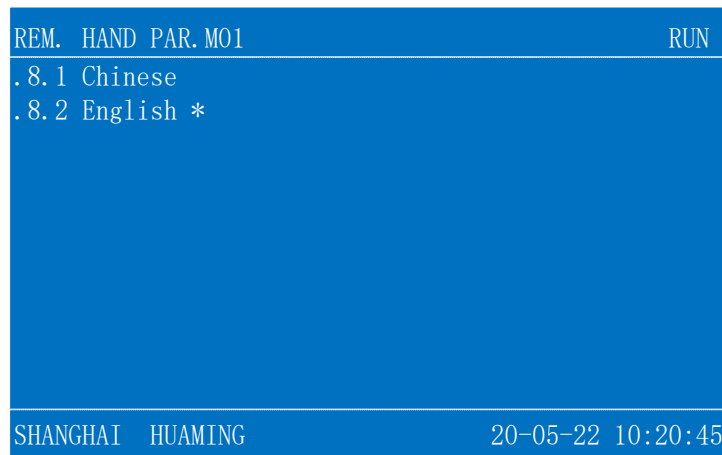
Import Constant Val : Pass a file through FTP or IEC61850 , then import the Constant Val.

Export Constant Val : Export the Constant Val, then export files over the FTP or IEC61850.

Default Setting : **All settings will be reset to default value.**

Update firmware: After clicking on the update firmware, the new and old firmware versions will be displayed. After confirmation, enter the password to start the upgrade. During the process, do not operate the device or power off.

4.10 Language View



User can switch HMI language in this page, the operation need to input the system password to be confirmed.

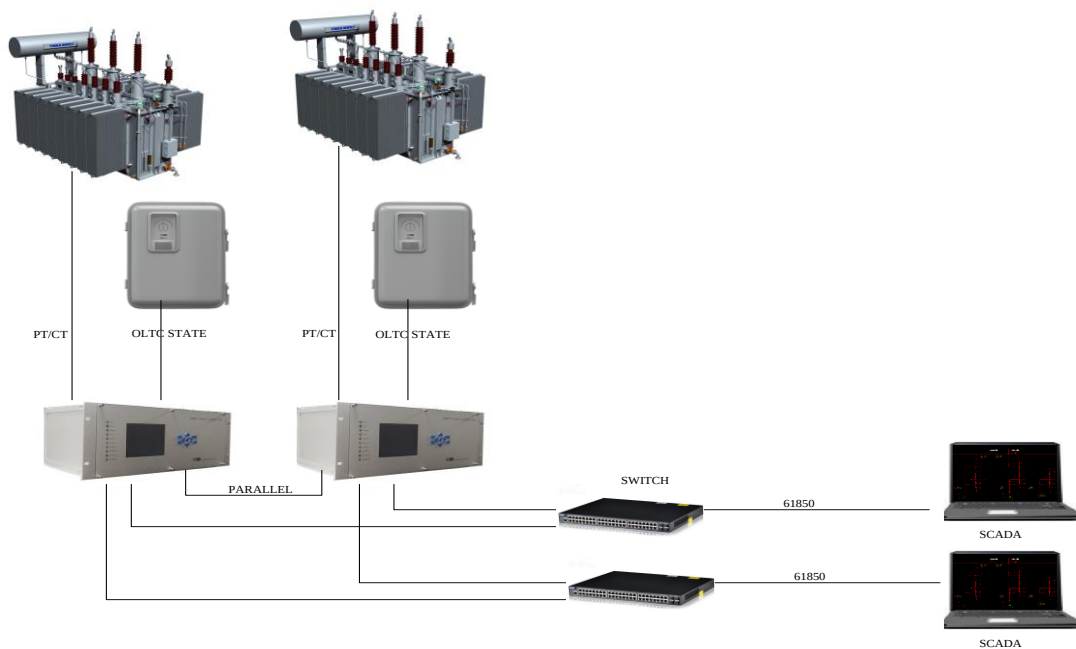
4.11 LED Description

LED Name	On	Off	Flash
Run	Run Mode	N/A	Debug Mode
Fault	AVR Faulty	OK	N/A
Remote	Remote Mode	Local Mode	Error State

Auto	Auto Mode	Manual Mode	Error State
Parallel	Parallel Mode	Independent Mode	N/A
Par.Block	Parallel Block	OK	N/A
UI Block	Voltage or Current Block	OK	N/A
TapBlock	OLTC Block	OK	AVR & OLTC Communication Error
AVR Failed	Tap change failed	OK	N/A
Regulating	Raising or Lowering	OK	N/A

5. Application

5.1 Paralleled AVR Diagram



6. Appendix

6.1 Appendix A: IEC 61850

IED	LD	LN	DO	DA	CDC	FC	Description
HMIET001	LD0	ATCC1	CtlV	mag.f	MV	MX	Control Voltage
HMIET001	LD0	ATCC1	LodA	mag.f	MV	MX	Load Current (total transformer secondary current)
HMIET001	LD0	ATCC1	PhaDiff	mag.f	MV	MX	Phase Difference
HMIET001	LD0	ATCC1	SecIa	mag.f	MV	MX	Secondary active current
HMIET001	LD0	ATCC1	SecIr	mag.f	MV	MX	Secondary reactive current
HMIET001	LD0	ATCC1	Freq	mag.f	MV	MX	Frequency
HMIET001	LD0	ATCC1	PriValV	mag.f	MV	MX	Primary Voltage Value
HMIET001	LD0	ATCC1	PriValA	mag.f	MV	MX	Primary Current Value
HMIET001	LD0	ATCC1	PriIa	mag.f	MV	MX	Primary active current
HMIET001	LD0	ATCC1	PriIr	mag.f	MV	MX	Primary reactive current
HMIET001	LD0	ATCC1	TapPos	stVal	INS	ST	Tap Position
HMIET001	LD0	ATCC1	LastTapPos	stVal	INS	ST	Last Tap Position
HMIET001	LD0	ATCC1	OpCnt	stVal	INS	ST	Operation counter
HMIET001	LD0	ATCC1	RefVoltage1	Oper.ctlval	SPC	CO	Desired value 1
HMIET001	LD0	ATCC1	RefVoltage2	Oper.ctlval	SPC	CO	Desired value 2
HMIET001	LD0	ATCC1	RefVoltage3	Oper.ctlval	SPC	CO	Desired value 3
HMIET001	LD0	ATCC1	TapRise	Oper.ctlval	SPC	CO	Tap position rise
HMIET001	LD0	ATCC1	TapDown	Oper.ctlval	SPC	CO	Tap position down
HMIET001	LD0	ATCC1	TapStop	Oper.ctlval	SPC	CO	Tap position stop
HMIET001	LD0	ATCC1	AutoChg	stVal	DPC	ST	Auto/Manual operation
HMIET001	LD0	ATCC1	ParOp	stVal	DPC	ST	Parallel/Independent operation
HMIET001	LD0	ATCC1	MasterChg	stVal	DPC	ST	Parallel Master/Follower operation
HMIET001	LD0	ATCC1	RunMode	stVal	DPC	ST	Debug/Run Mode
HMIET001	LD0	ATCC1	TapChg	valWTr.pos Val	BSC	ST	Change Tap Position
HMIET001	LD0	ATCC1	AVRCommFault	stVal	SPS	ST	AVR communicate status
HMIET001	LD0	ATCC1	OltcGenBlock	stVal	SPS	ST	OLTC general block
HMIET001	LD0	ATCC1	OltcCommFault	stVal	SPS	ST	OLTC communicate status
HMIET001	LD0	ATCC1	AVRFaulty	stVal	SPS	ST	AVR Device fault
HMIET001	LD0	ATCC1	OverCurrBlock	stVal	SPS	ST	AVR over current block
HMIET001	LD0	ATCC1	OverVoltBlock	stVal	SPS	ST	AVR over voltage block
HMIET001	LD0	ATCC1	UnderVoltBlock	stVal	SPS	ST	AVR under voltage block
HMIET001	LD0	ATCC1	ParLostConn	stVal	SPS	ST	Parallel connection lost
HMIET001	LD0	ATCC1	ParSetErr	stVal	SPS	ST	Parallel setting error
HMIET001	LD0	ATCC1	Rasing	stVal	SPS	ST	OLTC tap rasing
HMIET001	LD0	ATCC1	Lowering	stVal	SPS	ST	OLTC tap lowering
HMIET001	LD0	ATCC1	Loc	stVal	SPS	ST	Local operation
HMIET001	LD0	ATCC1	TapInComplete	stVal	SPS	ST	TAP Changer In Complete
HMIET001	LD0	ATCC1	TapInProgress	stVal	SPS	ST	TAP Changer In Progress
HMIET001	LD0	ATCC1	TapOutStep	stVal	SPS	ST	TAP Changer Out of Step
HMIET001	LD0	ATCC1	EndPosR	stVal	SPS	ST	End position raise reached
HMIET001	LD0	ATCC1	EndPosL	stVal	SPS	ST	End position lower reached
HMIET001	LD0	ATCC1	PriVoltage	setMag.f	ASG	SG	Primary Voltage
HMIET001	LD0	ATCC1	SecVoltage	setMag.f	ASG	SG	Secondary Voltage
HMIET001	LD0	ATCC1	PriCurrent	setMag.f	ASG	SG	Primary Current
HMIET001	LD0	ATCC1	SecCurrent	setMag.f	ASG	SG	Secondary Current
HMIET001	LD0	ATCC1	LDCR	setMag.f	ASG	SG	Line drop voltage due to line resistance component
HMIET001	LD0	ATCC1	LDCX	setMag.f	ASG	SG	Line drop voltage due to line reactance component
HMIET001	LD0	ATCC1	LDCZ	setMag.f	ASG	SG	Line drop voltage due to line total impedance
HMIET001	LD0	ATCC1	BndCtr1	setMag.f	ASG	SG	Band center voltage 1 (FPF presumed)
HMIET001	LD0	ATCC1	BndCtr2	setMag.f	ASG	SG	Band center voltage 2 (FPF presumed)

HMIET - I Automatic Voltage Regulator

HMIET001	LD0	ATCC1	BndCtr3	setMag.f	ASG	SG	Band center voltage 3 (FPF presumed)
HMIET001	LD0	ATCC1	BndWid	setMag.f	ASG	SG	Band width voltage (as voltage or percent of nominal voltage, FPF presumed)
HMIET001	LD0	ATCC1	LimLodA	setMag.f	ASG	SG	Limit Load Current (LTC Block Load Current)
HMIET001	LD0	ATCC1	LimLodATime	setMag.f	ASG	SG	Limit Load Current Time
HMIET001	LD0	ATCC1	LimLodAVal	setMag.f	ASG	SG	Limit Load Current Block Return Value
HMIET001	LD0	ATCC1	LimLodV	setMag.f	ASG	SG	Limit Load Voltage(LTC Block Load Voltage)
HMIET001	LD0	ATCC1	LimLodVTime	setMag.f	ASG	SG	Limit Load Voltage Time
HMIET001	LD0	ATCC1	LimLodVVal	setMag.f	ASG	SG	Limit Load Voltage Block Return Value
HMIET001	LD0	ATCC1	LowLimLodV	setMag.f	ASG	SG	Low Limit Load Voltage(LTC Block Load Voltage)
HMIET001	LD0	ATCC1	LowLimLodVTime	setMag.f	ASG	SG	Low Limit Load Voltage Time
HMIET001	LD0	ATCC1	LowLimLodVVal	setMag.f	ASG	SG	Low Limit Load Voltage Block Return Value
HMIET001	LD0	ATCC1	PhaseAngle	setVal	ING	SG	Phase Angle
HMIET001	LD0	ATCC1	CtIDITmmsT1	setVal	ING	SG	Control intentional time delay 1 (FPF presumed)
HMIET001	LD0	ATCC1	CtIDITmmsT2	setVal	ING	SG	Control intentional time delay 2 (FPF presumed)
HMIET001	LD0	ATCC1	AVRDirection	setVal	ING	SG	AVR Direction
HMIET001	LD0	ATCC1	LimLodABlk	setVal	ING	SG	Limit Load Current Blocking
HMIET001	LD0	ATCC1	LimLodVBlk	setVal	ING	SG	Limit Load Voltage Blocking
HMIET001	LD0	ATCC1	LowLimLodVBlk	setVal	ING	SG	Low Limit Load Voltage Blocking
HMIET001	LD0	ATCC1	LDC	setVal	ING	SG	Line Drop Compensation is R and X or Z model
HMIET001	LD0	ATCC1	PhaseAngle	setVal	ING	SG	Phase Angle

6.2 Appendix B: Modbus Rtu

Type	FuncCode	Addr	Name	Description	Remarks
DI	0x02	0	AVRCommFault	AVR communicate status	
		1	OltcGenBlock	OLTC general block	
		2	OltcCommFault	OLTC communicate status	
		3	AVRFaulty	AVR Device fault	
		4	OverCurrBlock	AVR over current block	
		5	OverVoltBlock	AVR over voltage block	
		6	UnderVoltBlock	AVR under voltage block	
		7	ParLostConn	Parallel connection lost	
		8	ParSetErr	Parallel setting error	
		9	Rasing	OLTC tap rasing	
		10	Lowering	OLTC tap lowering	
		11	Loc	Local operation	
		12	TapInComplete	TAP Changer In Complete	
		13	TapInProgress	TAP Changer In Progress	
		14	TapOutStep	TAP Changer Out of Step	
		15	EndPosR	End position raise reached	
		16	EndPosL	End position lower reached	
DO	0x01 /0x06	0	AutoChg	Auto/Manual operation	0:Auto 1:Manual
		1	ParOp	Parallel/Independent operation	0:Parallel 1:Independent
		2	MasterChg	Parallel Master/Follower operation	0:Master 1:Follower
		3	RunMode	Debug/Run Mode	0:Debug 1:Run
		4	CirMode	Circulating current mode	
		5	RefVoltage1	Desired value 1	
		6	RefVoltage2	Desired value 2	
		7	RefVoltage3	Desired value 3	
		8	TapRise	Tap position rise	
		9	TapDown	Tap position down	
		10	TapStop	Tap position stop	

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AI	0x04	0	TapPos	Tap Position	Float HH HL LH LL
		2	LastTapPos	Last Tap Position	
		4	CtIV	Control Voltage	
		6	LodA	Load Current (total transformer secondary current)	
		8	PriValV	Primary Voltage Value	
		10	PriValA	Primary Current Value	
		12	OpCnt	Operation counter	
		14	PhaDiff	Phase Difference	
		16	SecIa	Secondary active current	
		18	SecIr	Secondary reactive current	
		20	Freq	Frequency	
		22	PriIa	Primary active current	
		24	PriIr	Primary reactive current	
		26	CircA	Circulating current	
AO	0x03 /0x10	0	PriVoltage	Primary Voltage	Float HH HL LH LL Must be modified in debug mode.
		2	SecVoltage	Secondary Voltage	
		4	PriCurrent	Primary Current	
		6	SecCurrent	Secondary Current	
		8	PhaseAngle	Phase Angle	
		10	LDC	Line Drop Compensation is R and X or Z model	
		12	LDCR	Line drop voltage due to line resistance component	
		14	LDCX	Line drop voltage due to line reactance component	
		16	LDCZ	Line drop voltage due to line total impedance	
		18	BndCtr1	Band center voltage 1 (FPF presumed)	
		20	BndCtr2	Band center voltage 2 (FPF presumed)	
		22	BndCtr3	Band center voltage 3 (FPF presumed)	
		24	BndWid	Band width voltage (as voltage or percent of nominal voltage, FPF presumed)	
		26	CtIDITmmsT1	Control intentional time delay 1 (FPF presumed)	
		28	CtIDITmmsT2	Control intentional time delay 2 (FPF presumed)	
		30	AVRDirection	AVR Direction	
		32	LimLodA	Limit Load Current (LTC Block Load Current)	
		34	LimLodATime	Limit Load Current Time	
		36	LimLodAVal	Limit Load Current Block Return Value	
		38	LimLodABlk	Limit Load Current Blocking	
		40	LimLodV	Limit Load Voltage(LTC Block Load Voltage)	
		42	LimLodVTime	Limit Load Voltage Time	
		44	LimLodVVal	Limit Load Voltage Block Return Value	
		46	LimLodVBlk	Limit Load Voltage Blocking	
		48	LowLimLodV	Low Limit Load Voltage(LTC Block Load Voltage)	
		50	LowLimLodVTime	Low Limit Load Voltage Time	
		52	LowLimLodVVal	Low Limit Load Voltage Block Return Value	
		54	LowLimLodVBlk	Low Limit Load Voltage Blocking	
		56	SetOpCnt	Set the operation counter value	
		58- 99	Space	Space	
		100	Year	Year	int32 HH HL LH LL Can be modified at any time.
		102	Month	Month	
		104	Day	Day	
		106	Hour	Hour	
		108	Min	Min	
		110	Sec	Sec	

Example :

Read DO	TX	01 01 00 00 00 0B 7D CD
	RX	01 01 02 0C 00 BC FC
Read DI	TX	01 02 00 00 00 11 B8 06
	RX	01 02 03 00 08 01 BE 4E

HMIET - I Automatic Voltage Regulator

Read AO (0-57)	TX	01 03 00 00 00 3A C5 D9
	RX	01 03 74 42 DC 00 00 42 C8 00 00 43 FA 00 00 3F 80 00 41 20 00 00 42 C8 00 00 42 DC 00 00 3F 80 00 00 42 20 00 00 41 20 00 00 3F 80 00 00 42 F0 00 00 3F 00 00 00 3F 80 00 00 3F 80 00 00 42 F0 00 00 3F 00 00 00 3F 80 00 00 3F 80 00 00 42 A0 00 00 3F 00 00 00 3F 80 00 00 00 00 00 00 40 C0 00 00 1D A9
	TX	01 03 00 64 00 0C 04 10
	RX	01 03 18 00 00 07 E5 00 00 00 0C 00 00 00 02 00 00 00 09 00 00 00 1A 00 00 00 32 89 2F
Read AI	TX	01 04 00 00 00 1C F1 C3
	RX	01 04 38 40 C0 00 00 40 E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 40 C0 00 F4 1E
Set Do	TX	01 05 00 00 FF 00 8C 3A
	RX	01 05 00 00 FF 00 8C 3A
Set AO	TX	01 10 00 00 00 02 04 00 00 04 49 30 99
	RX	01 10 00 00 00 02 41 C8